ISE 218: Fundamentals of Information Technology

Stony Brook Institute at Anhui University

Homework #6 (Unit 6)

Spring 2025

Assignment Due: May 23, 2025

Neatly write or type your answers to the following problems on a separate sheet of paper and submit a digital copy on Microsoft Teams by the due date. For any problems involving mathematical calculations, you must show all work to receive full credit. Simplify your answers as much as possible, giving a single number as your answer.

Please clearly print or type your full name and ID # on the first page of your submission.

1. [5 pts] On a large system, suppose we can upgrade a CPU to make it 150% faster for $8,000 or upgrade its disk drives for $5,000 to make them 50% faster.
   Processes spend 40% of their time running in the CPU and 60% of their time waiting for disk service. An upgrade of which component would offer the greater benefit for the lesser cost?

   Amdahl's Law:

$$S = \frac{1}{(1-f) + \frac{f}{k}}$$

   Disk upgrade:
   f=0.6,k=1.5

$$S = \frac{1}{(1-0.6) + \frac{0.6}{1.5}} = 1.25$$

   Cost per 1% improvement:

   $5000/25 = $200

   CPU upgrade:
   f=0.4,k=2.5

$$S = \frac{1}{(1-0.4) + \frac{0.4}{2.5}} \approx 1.316$$

Cost per 1% improvement:

$$\$8000/31.6 \approx \$253.16$$

Conclusion:
The disk upgrade is more cost-effective (lower cost per 1% performance gain).

2. [5 pts] Suppose a computer's processing load consists of 65% CPU activity and 35% disk activity. You can upgrade your disks for $5,000 to make them 1.5 times as fast as they are currently. Or, you can upgrade your CPU to make it 2.5 times as fast for $8,000. Provide a cost-benefit analysis.

Amdahl's Law:

$$\boldsymbol{S = \frac{1}{(1-f)+\frac{f}{k}}}$$

Disk upgrade:
f=0.35,k=1.5

$$\boldsymbol{S = \frac{1}{(1-0.35)+\frac{0.35}{1.5}} \approx 1.132}$$

Cost per 1% improvement:

$$\$5000/13.2 = \$378.79$$

CPU upgrade:
f=0.65,k=2.5

$$\boldsymbol{S = \frac{1}{(1-0.6)+\frac{0.6}{2.5}} \approx 1.639}$$

Cost per 1% improvement:

$$\$8000/63.9 \approx \$125.20$$

Conclusion:
The CPU upgrade is more cost-effective (lower cost per 1% performance gain).

3. [5 pts] What are the differences among Interrupt-Driven I/O, Memory-Mapped I/O, Direct Memory Access (DMA), and Channel I/O? At least provide 2 differences for each I/O.
Descriptive answer

Interrupt-Driven I/O

Requires CPU intervention for each I/O operation.

The CPU handles the data transfer between the device and memory.

Memory-Mapped I/O

I/O devices are mapped into the CPU's memory address space.

The CPU uses standard memory access instructions to communicate with devices.

DMA (Direct Memory Access)

A DMA controller handles data transfer directly between memory and the device, bypassing the CPU.

The CPU is only involved in initiating and terminating the transfer.

Channel I/O

Uses an independent I/O processor (channel) to control I/O operations.

It can handle multiple devices and complex operations without CPU intervention.

4. Suppose a disk drive has the following characteristics:

- 2 surfaces
- 8,383 tracks per surface
- 63 sectors per track
- 256 bytes/sector
- Track-to-track seek time of 8.5 milliseconds
- Rotational speed of 7,200 RPM.

a. [5 pts] What is the capacity of the drive in MB? Use base-2 units of measure, i.e., 1 GB = 2^{30} bytes.

Capacity=2×8,383×63×256=268,435,456 bytes≈257.88MB

b. [5 pts] What is the access time in ms?

Average rotational delay:

$$\frac{60 \times 1000}{7200 \times 2} = 4.17ms$$

Access Time=Seek Time+ Average Rotational Delay
=8.5ms+4.17ms
=12.67ms

5. [5 pts] What is the difference between SSD and Magnetic disk Technology?

Feature	**SSD**	**Magnetic disk**
Storage Medium	Flash memory	Spinning magnetic platters
Speed	Faster	Slower
Cost	Higher	Lower
Consumption	Lower	Higher